

HO CHI MINH TAN SON NHAT INTL, Viet Nam

WMO: 489000

Lat: 10.819N

Lon: 106.652E

Elev: 33

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	68.2	70.1	58.8	74.2	79.7	61.2	80.7	80.6	15.3	81.7	13.6	81.6	4.0	0	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	15.5	96.5	78.2	94.8	78.3	93.4	78.2	82.5	88.6	81.9	88.0	81.4	87.6	8.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.9	85.0	80.6	159.0	84.9	79.7	154.4	84.2	46.5	88.8	45.7	89.1	45.2	88.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.8	13.3	64.9	98.9	2.2	1.3	63.3	99.9	62.1	100.7	60.8	101.4	59.2	102.4		
(5)			60.9	84.3	2.5	1.5	59.1	85.4	57.6	86.2	56.2	87.1	54.4	88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	80.7	82.3	84.8	86.4	85.7	83.9	82.9	83.0	82.5	82.1	81.7	80.4		
	DBStd	2.84	2.57	2.17	1.98	2.10	2.61	2.10	1.94	1.94	1.94	1.93	2.19	2.68		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12055	951	905	1079	1093	1106	1016	1020	1022	975	994	952	942		
	CDD65	6580	486	485	614	643	641	566	555	557	525	529	502	477		
	CDH74	68971	4625	4914	7006	7902	7511	6140	5592	5679	5139	5044	4902	4517		
	CDH80	28130	1792	2060	3132	3820	3474	2497	2100	2103	1827	1795	1848	1682		
(14) Wind	WSAvg	6.3	5.3	6.5	7.6	7.3	6.3	6.6	7.3	7.6	6.2	4.9	5.0	4.9		
(15) Precipitation	PrecAvg	69.0	0.6	0.3	1.2	2.5	7.6	9.3	10.0	10.1	11.1	10.2	4.8	1.3		
	PrecMax	83.0	3.5	2.2	4.7	9.0	12.1	15.2	21.8	16.6	20.4	15.4	12.4	3.8		
	PrecMin	54.1	0.0	0.0	0.1	0.2	1.3	2.5	4.3	3.0	6.1	5.6	0.1	0.0		
	PrecStd	7.5	0.9	0.5	1.3	2.2	2.8	3.4	3.9	3.5	3.1	2.9	3.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	95.0	97.1	97.5	98.3	95.2	93.5	93.5	93.3	93.1	93.2	92.9		
		MCWB	74.8	74.3	75.9	78.0	79.5	80.3	80.0	80.0	79.9	78.1	76.9	75.2		
	2%	DB	91.5	93.1	95.0	95.9	96.3	93.3	91.8	91.7	91.5	91.2	91.4	91.2		
		MCWB	75.0	74.6	76.8	78.4	79.5	80.2	79.8	79.6	79.9	78.7	77.6	75.9		
	5%	DB	89.9	91.5	93.3	94.6	93.7	91.7	91.0	90.9	89.9	89.7	89.8	89.6		
		MCWB	74.8	74.8	76.6	78.6	79.8	79.9	79.6	79.5	79.4	78.7	77.6	75.8		
	10%	DB	88.2	89.8	91.7	93.1	92.0	90.0	89.4	89.4	88.4	88.1	88.4	87.9		
		MCWB	74.6	74.7	76.5	78.5	79.8	79.7	79.4	79.4	79.4	78.9	77.7	76.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	79.6	81.0	82.6	83.3	83.2	82.8	82.7	82.4	82.0	81.5	80.2		
		MCDB	86.7	88.2	89.6	90.1	89.8	88.8	88.0	88.6	88.5	87.2	87.2	86.3		
	2%	WB	78.6	78.3	79.9	81.6	82.2	82.2	81.7	81.6	81.5	81.0	80.2	79.2		
		MCDB	85.5	86.9	88.3	88.8	88.6	88.1	87.3	87.6	87.5	86.5	86.1	85.3		
	5%	WB	77.5	77.4	79.1	80.8	81.6	81.4	80.9	80.7	80.7	80.3	79.5	78.4		
		MCDB	84.7	85.9	87.4	87.8	88.1	87.5	86.7	86.9	86.6	85.7	85.3	84.5		
	10%	WB	76.5	76.5	78.3	80.1	80.9	80.7	80.2	80.0	80.0	79.6	78.8	77.6		
		MCDB	83.8	84.8	86.4	87.2	87.7	87.0	85.9	85.9	85.5	84.9	84.6	83.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	17.9	16.8	15.5	15.2	14.4	14.0	13.6	13.4	13.9	14.8	15.9		
		MCDBR	18.1	18.7	17.7	16.3	16.2	15.8	15.5	15.2	15.1	15.4	16.1	17.2		
	5% WB	MCWBR	6.0	5.7	5.0	4.7	5.2	5.8	6.0	5.6	5.8	5.8	5.7	6.0		
		MCWBR	15.8	16.5	15.8	15.1	15.2	14.5	14.0	13.7	13.7	13.9	14.6	14.9		
(40) Clear-Sky Solar Irradiance	taub	0.488	0.513	0.551	0.530	0.461	0.452	0.452	0.455	0.460	0.495	0.461	0.461	0.461		
		taud	2.157	2.078	2.012	2.093	2.335	2.382	2.372	2.356	2.331	2.219	2.314	2.276		
	Ebn at Noon	260	258	250	253	266	265	266	269	270	259	266	265	265		
		Edn at Noon	48	53	58	53	41	38	39	40	42	46	41	42		
(44) All-Sky Solar Radiation	RadAvg	1626	1864	1905	1878	1698	1590	1557	1561	1534	1527	1539	1492	1492		
	RadStd	87	81	89	90	114	66	85	109	104	122	93	120	120		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-1.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.40	+0.68	N/A	N/A	+0.40	+0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page